

Niyath Nair

Bengaluru, India · iamniyathnair@gmail.com · github.com/Niyathnair · linkedin.com/in/niyath-nair-9601b0231

EXPERIENCE

AntHill AI — Founding Research Engineer

Jul 2026 – Present

- Built a Claude-based retrieval agent combining BM25, vector and Personalized PageRank graph search, lifting internal enterprise-QA benchmark score from **48 to 90** (94% answer accuracy).
- Raised reranking **nDCG@10 from 0.75 to 0.97** with a bi-encoder reranker and served full-graph PageRank at **~15 ms/query** via sparse caching.
- Engineered a source-agnostic LLM knowledge-graph pipeline with entity resolution and time-aware facts, scaling to **82K+ entities and 146K+ relations** per production build across Slack, Jira, Confluence, GitHub and Drive; cut build time 40%.
- Developed an ontology-grounded text-to-SQL agent that reached **#2 on the LiveSQLBench-Base-Lite leaderboard** (50.4%, Sep 2026), ahead of o3-mini (42.6%) and GPT-4o (34.4%).
- Designed multi-tenant security and infrastructure: Postgres row-level security mirroring source-system permissions, service-wide RBAC and per-customer AWS stacks in Terraform.

slice Small Finance Bank — Product Analyst

May 2025 – Jul 2026

- Led automation of the bank's RBI regulatory reporting with a multi-agent pipeline that extracts, reconciles and validates data and prepares submission-ready returns, **saving 3 FTEs** of manual work.
- Built Donna, a Slack-native AI assistant connected via MCP to the bank's data and internal tools, answering questions across every business function, from analytics and reporting to marketing ideas; adopted by finance, analytics, marketing and other non-technical teams.

Jio Platforms

Data Science Intern

May – Jul 2024

- Fine-tuned transformer sentiment classifiers to **0.90 F1** and clustered 1.1M user reviews with DBSCAN.

Data Scientist Intern

May – Jul 2023

- Improved 5G signal-strength prediction by 8% (**0.912 F1, 0.94 ROC-AUC**) with XGBoost and path-loss diffraction features (EP-MKED).

PROJECTS

CLIF-GPT: teaching LLMs intuitive physics from video — PyTorch

github.com/Niyathnair/CLIF-GPT

- Built a model that gives a frozen LLM **physical common sense learned from video**: it tracks objects, learns cause and effect (support lost → fall) and steers next-token predictions on **unseen prompts, zero-shot**.
- Designed an object-centric dynamics encoder (slot attention, YOLO-World tracking, neural-ODE state prediction) fused into the LLM via cross-attention and a low-rank logit-bias projector, trained JEPA-style.
- Derived a **KL-divergence bound** between the fused and ideal causal next-token distributions; shipped an end-to-end Physion pipeline with live webcam and web demos.

Curvetopia — Adobe GenSolve Hackathon

github.com/Niyathnair/Adobe-Gensolve

- Built a geometry pipeline that identifies, regularizes and completes hand-drawn 2D curves: shape-context matching with Hungarian assignment to classify shapes, PCA-aligned fitting, symmetry detection via reflected-point matching and spline merging of fragmented, occluded contours.

Generative stress testing for portfolio risk — IIT Jodhpur, TensorFlow

github.com/Niyathnair/Financial_Risk_Assesment

- Trained a Bi-GAN to generate liquidity and volatility shock scenarios and estimated 95% VaR and CVaR from them, capturing worst-case losses better than a GARCH baseline.

EDUCATION

Indian Institute of Technology Jodhpur — B.Tech

2021 – 2025

Core Member, Google Developer Student Club (2022–25)

AWARDS

Powerhouse Player Award, slice Small Finance Bank

2026

National Semi-Finalist, Tata Imagination Challenge

2025

SKILLS

Languages Python, SQL, TypeScript, C++

ML PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn, XGBoost

Multimodal SigLIP, YOLO-World, ByteTrack, slot attention, JEPA

LLMs & retrieval Agents, MCP, RAG, knowledge graphs, BM25, pgvector, reranking

Evals Test-set design, LLM-as-judge, recall@k / nDCG, ablations

Systems Postgres (RLS, asyncpg), FastAPI, Celery / Redis, Next.js, AWS (ECS, Aurora), Terraform